

Train Ticket Agent — Build Status & Remaining Work

Anmol · 2026-08-29 · [check/train-ticket-agent/](#)

What was asked: a multi-agent system that takes a natural-language train journey request, searches/compares IRCTC, ixigo and ConfirmTkt, schedules itself around Tatkal-style booking windows, and books the ticket — using Gemini 2.5 Flash, a web interface, and proper guardrails/security. This document is the short version of what actually got built, what was deliberately left out and why, and what's left for you to do.

What's built and verified working

- Planning docs: PRD.md, FRD.md, AUTOMATION_LIMITS.md (docs/ folder)
- FastAPI backend + SQLite, LangGraph orchestrator fanning out to 3 search agents in parallel and merging results
- Gemini 2.5 Flash: parses free-text requests into structured fields, writes plain-language result summaries
- Comparison/ranking agent, including a berth-type preference that boosts matching options
- APScheduler Tatkal-window scheduler — persistent job store, rejects trigger times already in the past, handles a missed fire (app asleep) by marking the job failed and notifying instead of silently dropping it
- Human-confirmation gate: short-lived single-use tokens, enforced twice independently (booking agent + guardrail), plus per-platform rate limiting on both search and staging
- Full audit log of every agent action
- Notifications: in-app banner + optional email via your own SMTP account
- Web UI: live progressive search status, berth-count display, scheduling with a live countdown, auto-refreshing jobs tab, inline confirm panel, passenger CRUD, audit log view
- Real train numbers/names can be pulled from India's open government data portal (data.gov.in) when configured — optional, falls back to a static list otherwise

Deliberately not built — a hard boundary, not a gap

No automated login to IRCTC/ixigo/ConfirmTkt, no CAPTCHA solving/bypass, no automated seat selection or payment submission, and no storage of IRCTC/platform credentials anywhere in the project. IRCTC's own Terms & Conditions state automation software is strictly prohibited, Section 143 of the Railways Act criminalizes procuring tickets via unauthorized software, and ixigo's robots.txt disallows scraping its search-result pages. The only legitimate path to real automation is IRCTC's official Authorized Agent API registration — see docs/AUTOMATION_LIMITS.md for the full sourced explanation.

What's left to build / your action items

Item	What it takes	Status
Real search data (train numbers/names)	Free data.gov.in signup + resource ID in .env	Optional, ~5 min
Real email notifications	Your own email's SMTP app-password in .env	Optional, ~5 min
Real IRCTC booking automation	IRCTC's official Authorized Agent registration (PAN+Aadhaar+agreement)	Your call — a real business/legal process, not a code change
Live search adapters for ixigo/ConfirmTkt	Blocked until each platform's own ToS/robots.txt is re-checked or an official API is found	Not started

Item	What it takes	Status
Auth / access control on the web app	Currently open on localhost, CORS *	Needed before exposing beyond your own machine
Secrets vault	Currently a plain .env file	Fine for solo local use; revisit if ever shared
Multi-passenger / multi-route batch scheduling	FRD Phase 3 scope	Not started
Alternate-route/date suggestions on no availability	FRD Phase 3 scope	Not started

How to run it

cd train-ticket-agent/backend, then: `..\..\venv\Scripts\python.exe -m pip install -r requirements.txt`, then: `..\..\venv\Scripts\python.exe -m uvicorn app.main:app --reload --port 8000`. Full details, including the .env setup, are in README.md.